

Q1) Write a function to find the 'longest Common prefix' string amongst an array of strings.

soln

```
def longest-common-prefix(strs):
```

```
    if not strs:
```

```
        return ""
```

```
    prefix = strs[0]
```

```
    for word in strs[1:]:
```

```
        while not word.startswith(prefix):
```

```
            prefix = prefix[:-1]
```

```
        if prefix == "":
```

```
            return ""
```

```
    return prefix
```

```
print(longest-common-prefix(["flower", "flow", "flight"]))
```

```
print(longest-common-prefix(["dog", "racecar", "car"]))
```

Q2) you are given a 0-indexed array of integers `nums` of length `n`. you are initially positioned at index 0.

Each element `nums[i]` represents the maximum length of a forward jump from index `i`. In other words if you are at index `i`, you can jump to any index `i + j` where

\*  $0 \leq j \leq \text{nums}[i]$  and

\*  $i + j < n$

Return the minimum number of jumps to reach index `n-1`. The test cases are generated such that you can reach index `n-1`.

soln

```
def jump(nums):
```

```
    jumps = 0
```

```
    current_end = 0
```

```
    farthest = 0
```

```
    for i in range(len(nums)-1):
```

```
        farthest = max(farthest, i + nums[i])
```

```
        if i == current_end:
```

```
            jumps += 1
```

```
            current_end = farthest
```

```
    return jumps
```

```
print(jump([2,3,1,1,4]))
```

```
print(jump([2,3,0,1,4]))
```